

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering ASSESSMENT TOOL 1 – SHORT ANSWER TEST

|                  |                                                                                                                                 |               |                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------|
| Course Code:     | CSA16                                                                                                                           | Course Title: | Data Warehousing and Data Mining      |
| CO Assessed:     | CO1 – Precision (BL1, BL2)                                                                                                      | Assessment:   | Assessment Tool 1 – Short Answer Test |
| Weightage:       | 10% of CIA                                                                                                                      | Total Marks:  | 100 (5 Questions × 20 Marks)          |
| CO1 Statement:   | <i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i> |               |                                       |
| Student Name:    | Ch. Sathish                                                                                                                     | Reg. No.:     | 192311032                             |
| Section / Batch: | CSE                                                                                                                             | Date:         | 17/7/26                               |

### Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

| Question Type     | Focus Area                                | Questions | Marks                                          |
|-------------------|-------------------------------------------|-----------|------------------------------------------------|
| Short Answer Test | Architecture of Data Warehouse, ETL, OLAP | Q1 – Q5   | Q1 –20<br>Q2 –20<br>Q3 –20<br>Q4 –20<br>Q5 –20 |
| GRAND TOTAL:      |                                           |           | 100 Marks                                      |

### SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit\_count and avg\_charge. At the lowest conceptual level, avg\_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?



2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction\_Fact and Loan\_Fact, sharing dimensions customer, branch, time, and account\_type.

- Transaction\_Fact → transaction\_count, amount
- Loan\_Fact → loan\_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I ch. sathish (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ch. sathish

## RUBRICS FOR CALCULATION

| Criteria                  | Wt. | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25% | Demonstrates thorough, accurate understanding of concepts   | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25% | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15% | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |
| Accuracy                  | 10% | Completely accurate and covers all required points          | Mostly accurate with minor omissions    | Partially correct with missing points        | Incorrect answers with major gaps        |
| Clarity                   | 5%  | Clear, neat, professional presentation                      | Understandable with minor issues        | Limited clarity, readability issues          | Poorly written, difficult to understand  |

### Marks Evaluation:

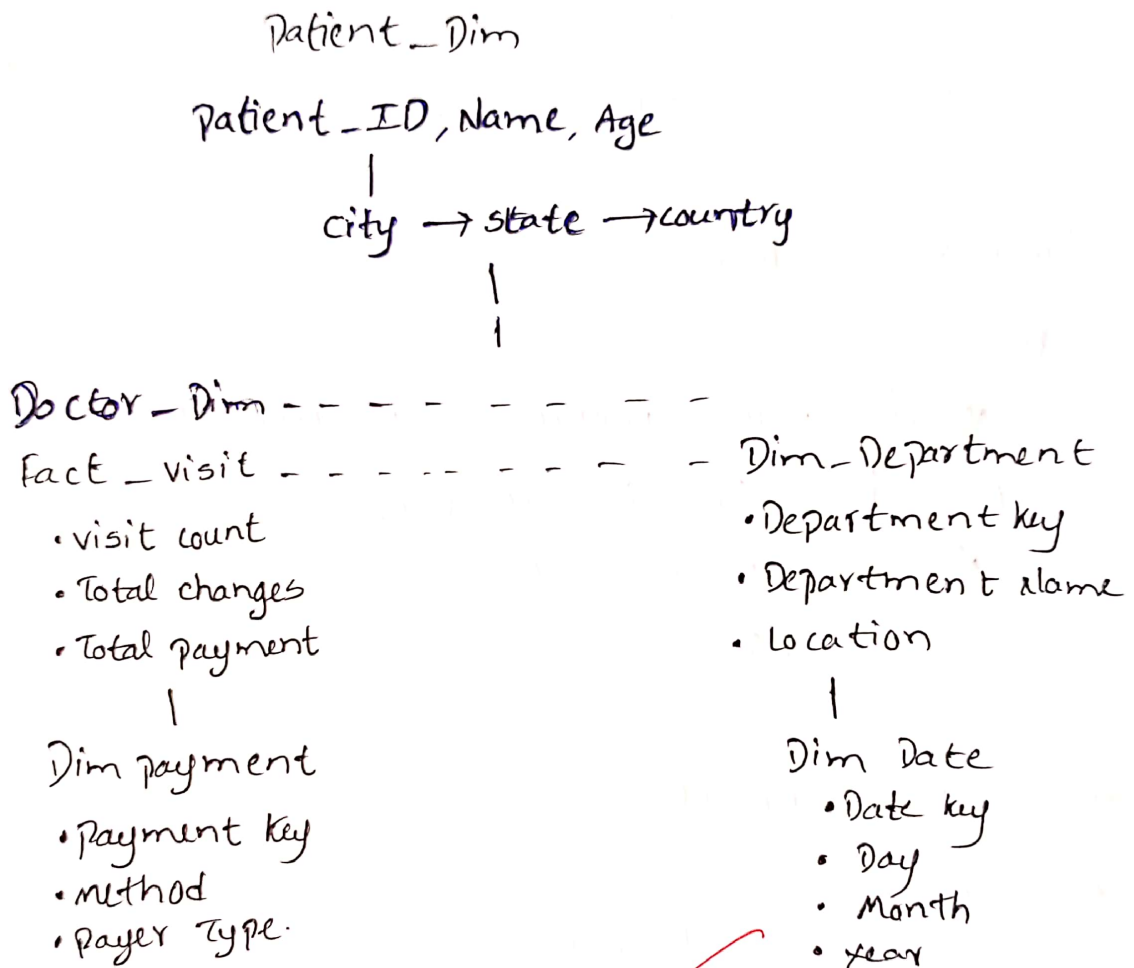
| Question No.        | Understanding of Concepts (25%) | Explanation (25%) | Application (20%) | Organization (15%) | Accuracy (10%) | Clarity (5%) | Total Marks |
|---------------------|---------------------------------|-------------------|-------------------|--------------------|----------------|--------------|-------------|
| Q1<br>(20 Marks)    |                                 |                   |                   |                    |                |              |             |
| Q2<br>(20 Marks)    |                                 |                   |                   |                    |                |              |             |
| Q3<br>(20 Marks)    |                                 |                   |                   |                    |                |              |             |
| Q4<br>(20 Marks)    |                                 |                   |                   |                    |                |              |             |
| Q5<br>(20 Marks)    |                                 |                   |                   |                    |                |              |             |
| Overall Total (100) |                                 |                   |                   |                    |                |              |             |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## Data warehousing and Data Mining

① city Hospital Data warehouse

Ans a) Snowflake schema Diagram



b) OLAP operations

starting cuboid: [patient, Doctor, Department, Time]

operations:

- i) Roll-up patient → All
- ii) Roll-up Doctor → All
- iii) Roll-up Time from Day → month → quarter → year
- iv) Keep Department dimensions unchanged.



Final cuboid: [All, All, Department, Year]

Output: Average consultation

| Department | Year | Fee  |
|------------|------|------|
| cardiology | 2025 | £850 |
| Neurology  | 2025 | £920 |

c) Number of cuboids

Formula

no. of cuboids = product of hierarchy levels

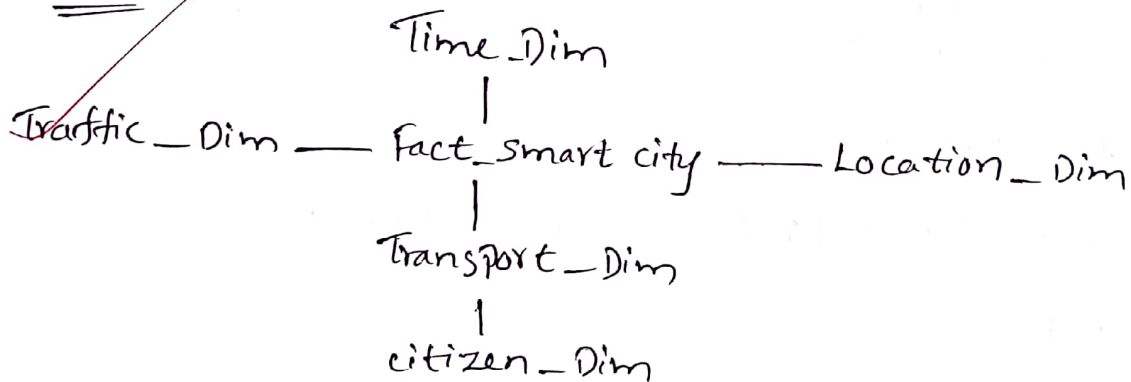
Each dimension has 4 levels

No. of dimensions = 4

$$4^4 = 256$$

② Smart city project

Star schema



Fact Table

Fact\_Smart city

Measures

- vehicle count
- congestion level

- Energy consumption
- Air Quality Index
- Noise level
- Ridership
- Punctuality
- Feedback store

### Dimensions

#### Traffic Dimensions

- sensor ID
- Road
- Zone

#### Location Dimension

- city
- Area
- Zone

#### citizen Dimension

- Age Group
- Feedback Type
- use category

### Justification

- Measures store numerical values for analysis.
- Dimensions provide different viewpoints for reporting and OLAP operations.

#### Time Dimension

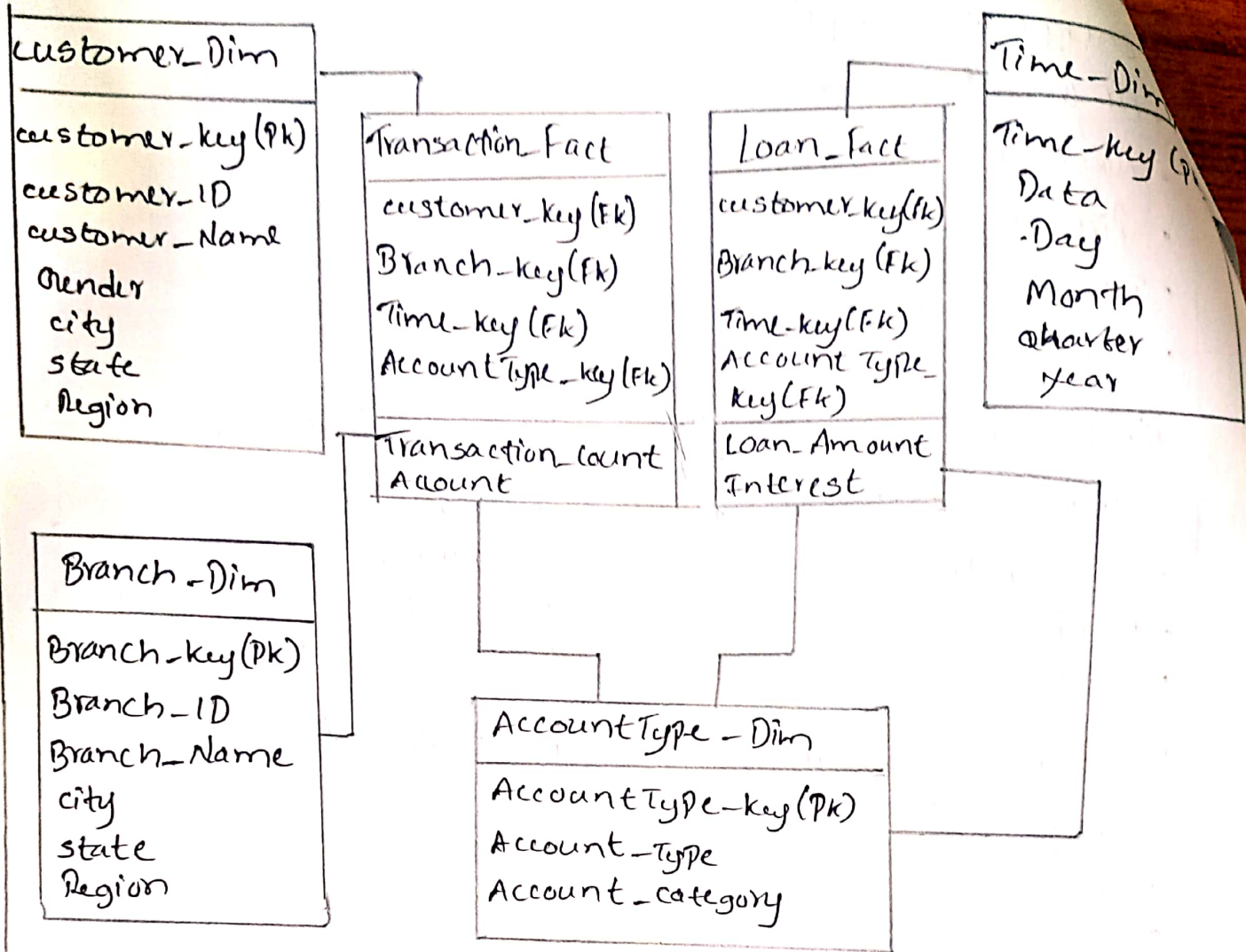
- year
- Quarter
- month
- Day

#### Transport Dimension

- Bus
- metro
- Route

## ③ Banking Data Warehouse

### a) Galaxy schema



## b) OLAP operations

Goal: Branch-wise yearly average transaction amount and loan amount.

### operations

- Roll-up Time → year
- Roll-up customer → All
- Roll-up Account Type → All
- keep Branch

Final cuboid: [Branch, year]

output:

| Branch  | year | Avg Transaction | Avg Loan  |
|---------|------|-----------------|-----------|
| chennai | 2025 | ₹18,000         | ₹5,20,000 |

# Manufacturing company

## i) Roll-up

Summarize defect rates by product category for the current year across all factories.

Result

| Product Category | Defect Rate |
|------------------|-------------|
| Electronics      | 2.5%        |
| Mechanical       | 1.8%        |

## ii) Drill-down

Focus on one factory

Analyze factory  $\rightarrow$  Department  $\rightarrow$  Product Type

Example

| Factory | Department | Product Type | Defect Rate |
|---------|------------|--------------|-------------|
| Chennai | Assembly   | Mobile       | 2.1%        |

## iii) Slice

Condition

Supplier Rating  $\geq 4$

Display

Production units.

Example

| Supplier | Rating | Producing units |
|----------|--------|-----------------|
| ABC Ltd  | 5      | 25,000          |

## iv) Dice

Conditions

selected product categories AND selected factory locations

Measures

• Defect Rate • Supplier Reliability.



## ⑤ Retail chain

### i) Roll-up

Summarize Inventory Turnover

By

• product category, last quarter, Across all stores

Example

| product category | Inventory Turnover |
|------------------|--------------------|
| Electronics      | 12                 |
| clothing         | 8                  |

### ii) Drill-down

Focus

Electronics → subcategory → sku

Example

| subcategory | sku  | sales Revenue |
|-------------|------|---------------|
| mobile      | M101 | ₹ 8,50,000    |
| Laptop      | L205 | ₹ 12,40,000   |

### iii) slice

condition

• Gender = Male

• Membership = Bronze

Measure

Total sales Revenue

Example

| store   | sales Revenue |
|---------|---------------|
| chennai | ₹ 25,00,000   |